

N. KAVITHA

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PROFESSIONAL SUMMARY

Highly analytical and results-driven **Software Developer** with a Master of Computer Applications (MCA) and a solid technical foundation in full-stack web engineering, deep learning, computer vision, bio-mathematical modeling, and network security analysis. Proficient in Python, JavaScript, Flask, FastAPI, React, and MySQL. Demonstrated ability to translate complex algorithms and mathematical models into production-ready web platforms, high-performance 3D WebGL visualizations, and real-time medical diagnostic tools with clean software architecture.

TECHNICAL SKILLS

Languages:	Python, JavaScript (ES6+), C/C++, SQL
Web & Frameworks:	Flask, FastAPI, React.js, Three.js (WebGL), HTML5/CSS3, RESTful APIs
ML & Vision:	Deep Learning (CNNs, VGG16), OpenCV, MediaPipe, Scikit-learn, NumPy, Pandas
Database & Tools:	MySQL, Git, GitHub, Recharts, JSON Data Management
Core Concepts:	Data Structures & Algorithms, Object-Oriented Programming (OOP), Software Architecture, Agile Methodology

PROJECTS

Heart Digital Twin: Medication Response Simulation & CDSS

2026

Tech Stack: FastAPI, React.js, Three.js (WebGL), Recharts, Python

- Architected a full-stack clinical decision support web platform using bio-mathematical kinetics to simulate patient-specific cardiac outcomes.
- Engineered a FastAPI simulation engine combining a One-Compartment PK model and Sigmoidal Hill E_{max} PD model with autonomic baroreflex compensation.
- Integrated an interactive Three.js 3D WebGL canvas to render real-time heart models scaling beat velocity dynamically based on computed cardiac output.
- Implemented robust patient onboarding modules, side-by-side trend analysis visualizers using Recharts, and automated clinical PDF report generation.

CyberScope – URL & Network Exposure Analyzer

2025 – 2026

Tech Stack: Python, Web Reconnaissance Tools, DNS/IP APIs, JavaScript, REST APIs

- Built a centralized web cybersecurity platform for passive, non-intrusive target exposure assessment and external web reconnaissance.
- Engineered multi-layer exposure correlation inspecting domain validation, DNS/IP resolution, HTTP response headers, scripts, and metadata.
- Designed an intuitive web dashboard organizing multi-tier network and infrastructure security findings into structured vulnerability reports.

Eye Disease Classification Using Deep Learning (Phase 2)

2025

Tech Stack: Python, TensorFlow/Keras (VGG16), OpenCV, Flask

- Engineered an automated medical image classification platform using VGG16 CNN architecture for multi-class ocular disease detection.
- Implemented an OpenCV image preprocessing pipeline for dynamic image resizing, intensity normalization, and noise filtering.
- Deployed trained neural network models inside a lightweight Flask web application, delivering instant, database-free diagnostic predictions.

Eye Disease Classification – Exploratory Analysis & Feasibility (Phase 1)

2023 – 2024

Tech Stack: Python, OpenCV, NumPy, Scikit-Learn

- Conducted exploratory data analysis, dataset preprocessing, and feature extraction algorithm benchmarking to establish foundational model pipelines.

PATENTS & PUBLICATIONS

- Published Indian Patent Application (Co-Inventor):** "An Adaptive Multimodal Communication and Classroom Participation System for Students with Disabilities" (App No: 202641107831 A, Published: Sept 2026). Developed software-based multimodal accessibility integration using gesture tracking, eye-gaze navigation, and customizable eye-blink Morse code.
- Best Idea Award:** Awarded Best Idea Award in the SIH Internal Hackathon 2025.

EDUCATION

- Master of Computer Applications (MCA) — 84%**
Hindusthan College of Arts and Science, Coimbatore, TN
- Bachelor of Computer Applications (BCA) — 91%**
Hindusthan College of Arts and Science, Coimbatore, TN
- Secondary School (12th Grade) — 69%**
The Matric Hr. Sec. School
- Secondary (10th Grade) — 85%**
Athena Global International School